

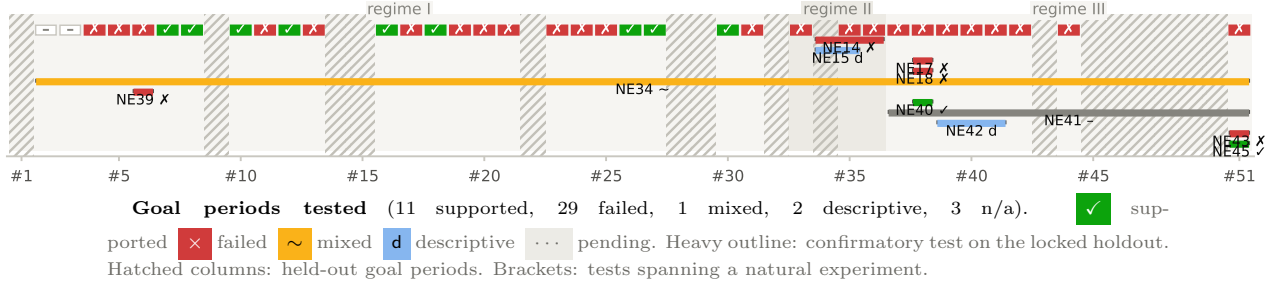
H36: A reorganization alarm: susceptibility and multi-information peak at transitions

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2.5/5

Bottom line. Multi-information and susceptibility barely mark reorganizations: on corrected data the alarm catches a quarter of goal changes, all via content, and a topic-shift detector does far better.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H36 computes three fluctuation statistics for each of 279 non-holdout days: multi-information, susceptibility and a heat-capacity analogue. Each uses 1-min activity spins, 4-state behavior and 30-min content vectors, as an excess over shuffled surrogates. The pre-registered alarm fires when the trailing z-score reaches 2. **The result is mixed.** The alarm catches 10 of 33 goal changes within a day, at a 10% placebo-window false-alarm rate (AUC 0.68, random-date $p = 0.03$). All of the signal is content: activity is at chance once roster and day-length changes are out, and ~40% of the content signal is within-day drift. The plain day-to-day shift of the mean content vector scores AUC 0.95. The alarm catches 1/13 scaffold changes. **Round 1b** uses a fixed activity table, removes restatements and adds a second embedding model. Hits fall to 0.21–0.27, and the random-date p rises to 0.08 (bge) and 0.21 (gte), at the failure boundary. Content (AUC 0.74–0.78) and topic shift (0.95–0.97) replicate. NE40 and NE45 go unseen. The holdout is not run.

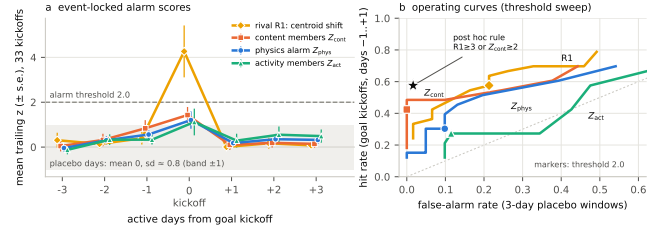
PHYSICAL MODEL

Models 01 (inverse Ising) and 11 (vector spins). The activity spin $s_i(t) = \pm 1$ marks agent i active in minute t ; $M = \sum_i s_i$. The content spin $\hat{s}_i(w)$ is agent i 's whitened unit content vector in window w . VR is the susceptibility as a variance ratio (1 for independent agents). c is a heat-capacity analogue with alignment energy E . I_G is the Gaussian multi-information (R : correlation matrix); $e(w)$ is the content alignment energy. Each statistic enters as observed minus surrogate mean, then as a trailing 10-day z .

$$\text{VR} = \frac{\text{Var}_t M}{\sum_i \text{Var}_t s_i}, \quad c = \frac{\text{Var}_t E}{n}, \quad E = -\frac{1}{n} \sum_{i < j} s_i s_j$$

$$I_G = -\frac{1}{2} \log \det R, \quad e(w) = -\langle \hat{s}_i(w) \cdot \hat{s}_j(w) \rangle_{i < j}$$

$$z_X(d) = \frac{X(d) - \text{med}_{10} X}{\text{sd}_{10}^{\text{trim}} X / c_{10}}, \quad Z_{\text{phys}} = \langle Z_I, Z_X, Z_C \rangle \geq 2$$



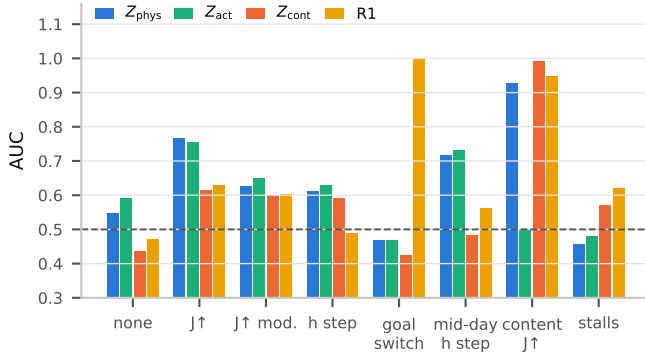
Observables. (a) Mean alarm scores around 33 goal kickoffs. The physics alarm (blue) and its content members (orange) rise on the kickoff day but stay below threshold on average. The activity members (aqua) barely move. The content-centroid shift (yellow) jumps. (b) Hit rate against placebo-window false-alarm rate as the threshold sweeps; markers sit at threshold 2. The star is the post hoc rule ($R1 \geq 3$ or $Z_{\text{cont}} \geq 2$): 58% hits at 2% false alarms, in-sample.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To flag a swarm reorganization, use a first-moment detector: the day-to-day shift of the mean content vector (AUC 0.95). Fluctuation statistics barely respond. Activity fluctuations follow who is present and for how long; content fluctuations follow the within-day settling after a new goal. Pending the holdout, the content-fluctuation score adds one use: a false-alarm filter beside the topic-shift detector. In round 1 the pair keeps the shift detector's hit rate at about a tenth of the false alarms. Scaffold changes, the events an operator most wants flagged, stay invisible to all of these daily statistics.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b bge / gte	Verdict
P1 goal hit $\leq 40\%$, hit 0.30 → 0.27 / 0.21; AUC 0.68 AUC < 0.65	→ 0.68 / 0.66; random-date p 0.03 → 0.08 / 0.21	mixed → mixed / failed
P2 content > activity	AUC 0.77 vs 0.55 → 0.78 / 0.74 vs 0.54 (trimmed 0.50)	held
P3 rival R1 wins	R1 0.95 → 0.95 / 0.97; Δ AUC -0.26 → -0.29 / -0.34	held
P5 scaffold at chance	1/13 → 1/13 / 2/13	held
P6 stalls cause false alarms	0 → 2/2 false alarms on top stall days (mask doesn't help)	failed → weak
P7 per-day 0.03-0.12	FAR 0.033 → 0.036 / 0.036	held
C3 rule $R1 \geq 3$ or $Z_{\text{cont}} \geq 2$	0.58 hits, 0.02 FAR → 0.55, 0.02 / 0.67, 0.13	model- dep.
NE39 chat (native) closed	content silent; activity alarm day +1 ($N = 4$)	failed
NE40, NE45 undoc-umented (native)	silent, as predicted: not detected blind	held
NE43b nudger (native) off	Z_{act} 3.1, R1 2.2 on day 0 (pre- dicted silent)	failed



Second figure. Synthetic validation at village sampling: 30 runs per scenario, AUC of the switch window against placebo windows. The statistics recover coupling switches in content, and in activity only partly. Day-boundary field steps and goal switches are invisible to the fluctuation statistics but obvious to the first-moment rival R1.

GOAL PERIODS IN DETAIL

G04 × 1b: failed (unchanged)
G05 × 1b: failed (unchanged)
G06 × 1b: failed (unchanged)
G07 ✓ 1b: supported (bge) / failed (gte)
G08 ✓ 1b: supported (unchanged)
G10 ✓ 1b: supported (unchanged)
G11 × 1b: failed (unchanged)
G12 ✓ 1b: supported (unchanged)
G13 × 1b: failed (unchanged)
G16 ✓ 1b: supported (unchanged)
+36 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The resolution is daily only. Round 1b has 55 placebo days (61 in round 1; the new NE dates exclude six), none in regime II. Most kickoffs fall on Mondays. The verdict sits on the random-date boundary and moves with the embedding model and the dedupe (bge without dedupe: $p = 0.11$). The activity channel reacts to roster, day length and, on the fixed table, joint silences. The operator rule is post hoc, and its false-alarm rate depends on the embedding model. The NE43b alarm is a single, unreplicated day. The synthetic figure uses round 1's generator.

ROUND 2

Re-freeze `confirm.py` on the fixed table, for both models, before any holdout run. Build HH268's combined monitor: topic shift (R1) plus H56's log features, H38's schedule signals and intraday windows.

- **Essence.** a cheap, log-computable alarm for structural change in a swarm (goal, roster, room, scaffold) with known hit and false-alarm rates; round 1 says content carries it, activity fluctuations don't, and a first-moment topic-shift detector beats the physics statistics.
- **H36-R1** Holdout confirmation.
- **H36-R2** Intraday alarm.
- **H36-R3** Sampling-invariant activity statistics.
- **H36-R4** Embedding swap.
- **H36-R5** Scaffold detector.

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